FABRIZIO PIETROBONO

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PROFILE

MSc student in Engineering in Computer Science and Artificial Intelligence at Sapienza University of Rome, with a BSc in Economics and Big Data from Roma Tre University. I combine a quantitative economics background with programming, data management and machine learning skills, built on university projects ranging from real-time distributed systems to data warehouses and the comparison of classification models. Working in hospitality, in the boating sector and in entertainment taught me to work in a team, handle customer relations and stay reliable under pressure.

EDUCATION

MSc in Engineering in Computer Science and Artificial Intelligence *Sep 2025 - present*
Sapienza University of Rome

- Exams passed: Machine Learning, Artificial Intelligence, Advanced Programming, Data Management, Algorithm Design, User-Driven Software Engineering, Dependable Distributed Systems, Economics and Computation, Interactive Graphics, Cybersecurity.

BSc in Economics and Big Data *16/10/2025*
Roma Tre University - Final grade: 96/110

High School Diploma in Sciences
Liceo Scientifico Giuseppe Peano, Rome

UNIVERSITY PROJECTS

E.C.H.O. - Distributed real-time seismic monitoring
Sapienza, Advanced Programming - team project (5 people)
GitHub: github.com/Biziopietrobono/echo-seismic-monitoring

- Microservice platform (FastAPI, WebSockets, Docker, PostgreSQL, Nginx, React) that ingests 20 Hz sensor streams and classifies earthquakes, explosions and nuclear-like events in real time.
- **My contribution:** the ingestion broker - sensor discovery, WebSocket ingestion, reconnection with exponential backoff, a dead-letter queue for invalid messages, and fan-out of the stream to the processing replicas.

Poker Hand Classification - Comparative study of classifiers
Sapienza, Machine Learning - individual project
GitHub: github.com/Biziopietrobono/poker-hand-classification

- Comparison of Naive Bayes, Softmax Regression, Decision Tree and Random Forest on over one million poker hands (10 highly imbalanced classes), tuned with Grid Search and stratified cross-validation.
- Best model: Random Forest, 75.7% accuracy and 0.98 micro-average AUC; analysis with confusion matrices, ROC curves, feature importance and learning curves.

The Purchasing Power of the Tech Worker - Data Warehouse
Sapienza, Data Management - team project (2 people)
GitHub: github.com/Biziopietrobono/tech-worker-purchasing-power-dw

- Integration of three heterogeneous sources (a 65k-response Stack Overflow survey, a country master and a cost-of-living index) into a PostgreSQL star schema with an idempotent Python ETL.
- OLAP analysis (ROLLUP, window functions, medians) and a Streamlit dashboard to find where a tech salary buys the most.

Penalty Kick - Interactive 3D game with Three.js and WebGL
Sapienza, Interactive Graphics - team project (3 people)
GitHub: github.com/SapienzaInteractiveGraphicsCourse/final-project-shootoutsquad | playable demo

- Lazio vs Roma penalty shootout playable in the browser, as shooter or goalkeeper, in training mode or a full match with five kicks per team and sudden death.

- Hierarchical shooter and goalkeeper models animated by hand, ball physics written from scratch (projectile motion), shadow-mapped lighting, colour, normal and specular textures, and a crowd rendered with InstancedMesh.

MyPiggyBank - User-centred design of a budgeting app

Sapienza, User-Driven Software Engineering - team project (5 people)

- 31-respondent user survey, requirements derived from it, dialogue modelled with state-transition networks, and a React prototype.

WORK EXPERIENCE

Bartender *2023/24 - present*

Nero Lab, Rome (EUR)

- Cocktail and drink preparation, bar and till management.
- Direct customer service, including at peak times.

Waiter *2021/22 - present*

Several restaurants in Rome, including Il Casale and Via Vai

- Welcoming guests, taking orders and table service.
- Coordinating with kitchen and floor staff to keep service fast and accurate.

Self-employed - Boating services *Summer 2026*

- Ran my own activity with boats: client management and organising the work independently.

Entertainment staff *Summer 2023*

- Organised and led entertainment and group activities.

SKILLS

Programming: Python, SQL, JavaScript

Data and machine learning: PostgreSQL, data warehousing (star schema), ETL, OLAP, scikit-learn, pandas

Tools: Git and GitHub, Docker, FastAPI, WebSockets, React, Streamlit, Three.js

Soft skills: Teamwork, customer relations, working independently

Languages: Italian (native), English (B2), Spanish (B1)

I authorise the processing of my personal data pursuant to Regulation (EU) 2016/679 (GDPR) and Italian Legislative Decree 196/2003.